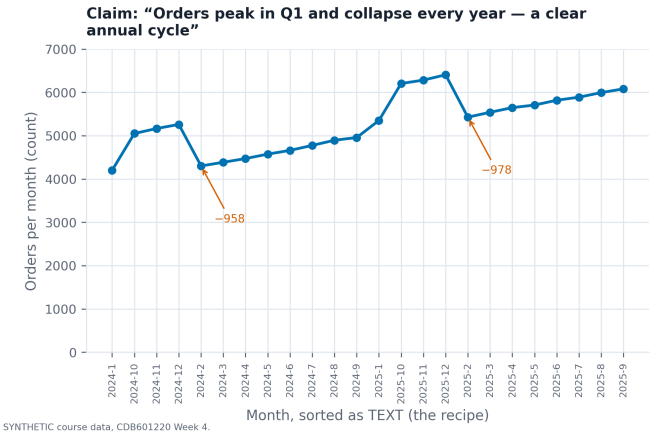
Week 4 · Part A — nine charts, rebuilt and taken apart

Areshay Asad · Roll 052 · BSAI · Section 7A · CDB601220 Data Visualization · all data SYNTHETIC course data

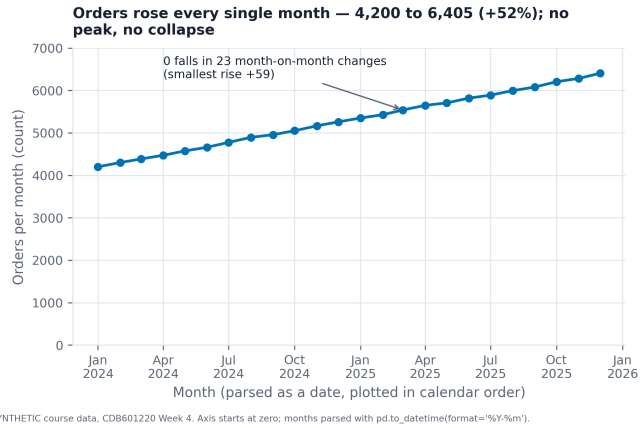
How to read each page. Both figures are shown (left: the analyst's chart rebuilt from the recipe; right: the honest redesign). Each case then gives: (1) the rebuild and its self-check, (2) the conclusion the chart invites *and* the first objection that does not hold, with evidence, (3) the mechanism with its number, (4) the redesign. Cases 5 and 9 are *incomplete* rather than wrong: their redesigns are the same chart with something else beside it.

Case 1 · “Orders peak in Q1 and collapse every year — a clear annual cycle”

data: case01_orders_stringdate.csv



Rebuilt (the claim)



Redesign (honest)

1 · Rebuild. Sorted the 24 rows by the text column **month**; 24 points; the line climbs, drops **958**, climbs, drops **978**, climbs again — self-check hit.

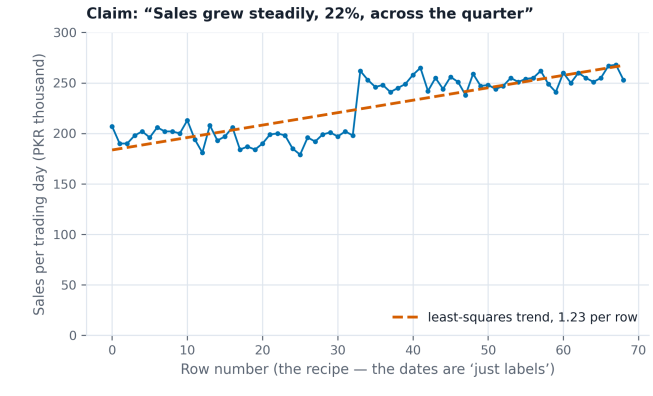
2 · Conclusion & decoy. Conclusion invited: orders have an annual cycle that peaks early in the year and then collapses. **Decoy:** “it is ordinary retail seasonality.” That does not hold: in calendar order there are **0 falls in 23 month-on-month changes** (the smallest rise is +59), so there is no peak for a season to produce. The series simply climbs 4,200 → 6,405 (+52%, ≈1.9% a month).

3 · Mechanism. month is text written *YYYY-M* with no zero padding, so a text sort puts “2024-10”, “2024-11”, “2024-12” *before* “2024-2”…“2024-9” (and the same in 2025). 22 of the 24 rows land in the wrong position. The “collapse” of **−958** is 2024-12 (5,259) followed by 2024-2 (4,301); the second, **−978**, is 2025-12 (6,405) followed by 2025-2 (5,427). Two-month gaps of ten months each.

4 · Redesign. Parse with *pd.to_datetime(format='%Y-%m')*, sort on the datetime, plot in calendar order, axis from zero.

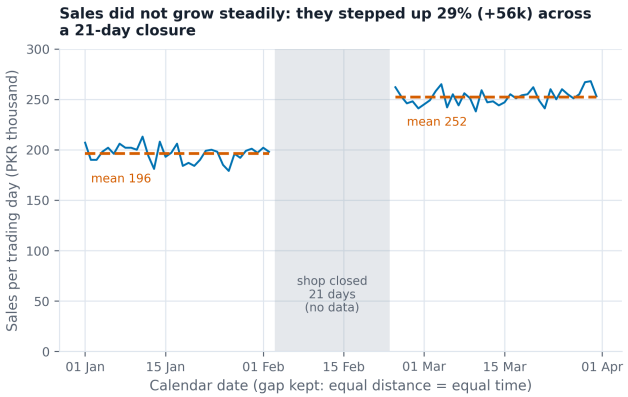
Case 2 · “Sales grew steadily, 22%, across the quarter”

data: case02_shop_closed.csv



SYNTHETIC course data, CDB601220 Week 4.

Rebuilt (the claim)



SYNTHETIC course data, CDB601220 Week 4. Days with no trading are NaN, so the line breaks.

Redesign (honest)

1 · Rebuild. Plotted **sales_pkr_k** against row number with a least-squares line: 69 points, first 207, last 253 (+22%); slope 1.23 per row — self-check hit.

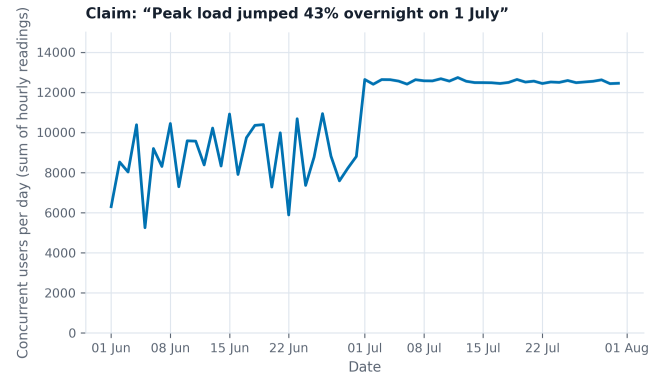
2 · Conclusion & decoy. Conclusion invited: steady growth. **Decoy:** “the rise is a day-of-week mix effect.” Ruled out: mean sales by weekday only span 222.4–231.9 (4.3% top to bottom) and every weekday has 9–10 observations, which cannot make a 29% step.

3 · Mechanism. Only trading days are recorded: 69 rows span 90 calendar days. There are **21 consecutive missing days (2025-02-03 to 2025-02-23)** — the shop was closed. Plotting against row number deletes the gap, so equal distance no longer means equal time and a one-day jump (198 → 262) is drawn as a ramp. Mean before the closure 196.2 (33 days), after 252.3 (36 days): a step of **+56.1 (+29%)**. Within each segment the trend is -0.16/day then +0.23/day — not steady growth. The row-number slope (1.23) also overstates the calendar slope (0.90/day). The data cannot say *why* sales stepped up on reopening.

4 · Redesign. Same data on a calendar axis, the closure kept as a shaded break (NaN, so the line breaks), and the two segment means shown.

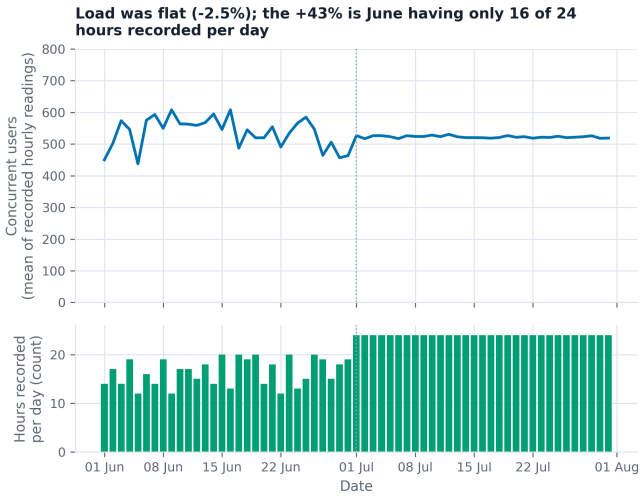
Case 3 · “Peak load jumped 43% overnight on 1 July”

data: case03_bandwidth_levels.csv



SYNTHETIC course data, CDB601220 Week 4.

Rebuilt (the claim)



SYNTHETIC course data, CDB601220 Week 4. A level (users connected at an instant) is averaged, never summed.

Redesign (honest)

1 · Rebuild. Resampled **concurrent_users** to daily with .sum(): June mean 8,782, July mean 12,543 (+43%) — self-check hit.

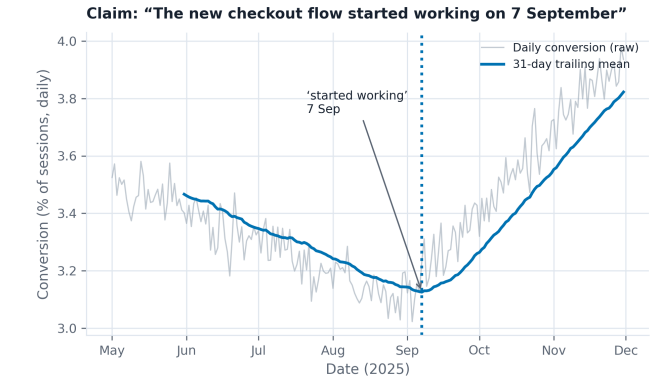
2 · Conclusion & decoy. Conclusion invited: traffic stepped up by 43% on 1 July. **Decoy:** “bytes_served also jumped 43%, so it is corroborated.” Ruled out: bytes are a flow, but June has only 16.4 of 24 hours recorded per day; per *recorded hour* bytes went 1.062M → 1.037M (-2.3%). The “corroboration” is the same artefact.

3 · Mechanism. concurrent_users is a *level* (users connected at an instant), and a level must never be summed over a day. June has 493 readings out of 720 possible hours (68.5%; 12–20 per day, never 24), July has all 744. The sum is mostly *how many rows the day has* (corr 0.97); hours per day rise 1.46× on 1 July, almost exactly the +43%. The honest aggregation is the mean: 535.9 → 522.6 (**-2.5%**). Even the daily maximum rose only +3.7% (779 → 808), and that is an upper bound because June's missing hours bias its maximum down.

4 · Redesign. Daily *mean* of recorded readings with, underneath, the hours recorded per day (a same-chart-plus-context fix; no dual axis).

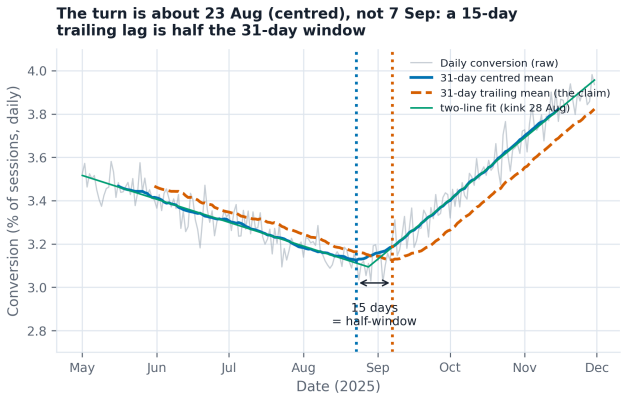
Case 4 · “The new checkout flow started working on 7 September”

data: case04_intervention_lag.csv



SYNTHETIC course data, CDB601220 Week 4.

Rebuilt (the claim)



SYNTHETIC course data, CDB601220 Week 4. y-axis does not start at zero (line chart of a rate, not bars).

Redesign (honest)

1 · Rebuild. Raw conversion in grey with a 31-day *trailing* mean; its minimum is 2025-09-07 — self-check hit.

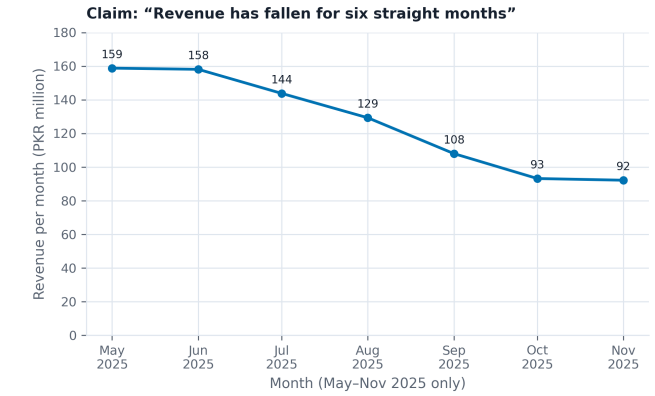
2 · Conclusion & decoy. Conclusion invited: the flow began to work on 7 Sep. **Decoy:** “the series is too noisy to date any turn, so the rise is just noise.” Ruled out: the day-to-day noise is ≈0.06 points, the rise is 3.12 → 3.88 (+0.76), and weekly means rise in 13 of 13 consecutive weeks after the trough. The turn is real; the *date* is what is wrong.

3 · Mechanism. A trailing mean at day t averages days t−30…t, so it cannot register a turn until about half the window is post-turn. Measured: centred 31-day minimum **2025-08-23**, trailing minimum **2025-09-07**, lag **15 days = half-window (15)**. Two independent checks agree with the centred date (7-day centred minimum 2025-08-26; a two-line fit puts the kink at 2025-08-28). So the turn happened 23–28 Aug, 10–15 days before the claimed date; 7 Sep is a property of the smoother. (The data cannot say when the checkout shipped.)

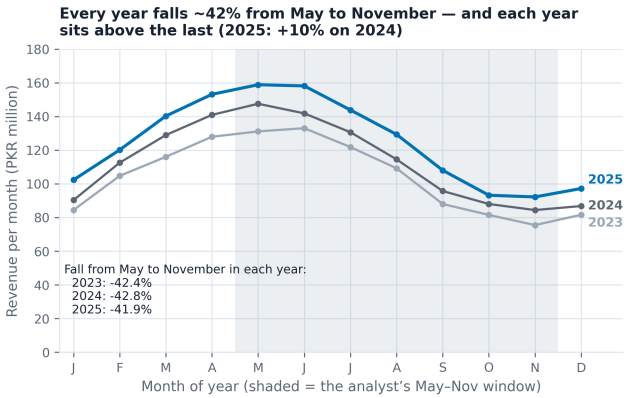
4 · Redesign. Raw, centred and trailing means together, both turning points marked, the lag drawn as an arrow, the two-line fit overlaid.

Case 5 · “Revenue has fallen for six straight months”

data: case05_revenue_window.csv



SYNTHETIC course data, CDB601220 Week 4.



SYNTHETIC course data, CDB601220 Week 4. Cycle plot: one line per year (tslib.cycle_plot idea).

Rebuilt (the claim) **Redesign (honest)**

1 · Rebuild. Took May–Nov 2025 only, line with markers, y from zero: 7 points, 158.8 → 92.2 (−41.9%) — self-check hit.

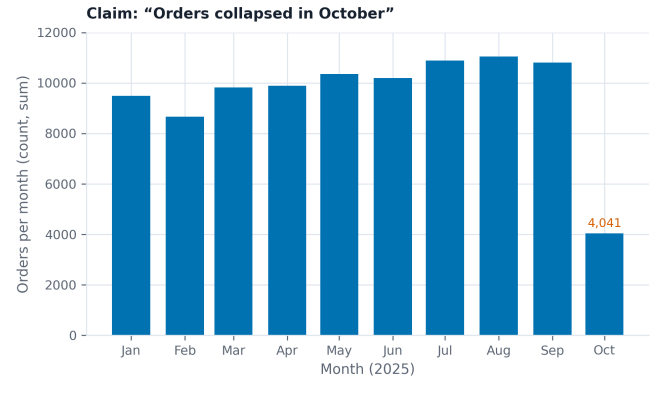
2 · Conclusion & decoy. Conclusion invited: the business is in decline. **Decoy (the two the brief warns about):** “the axis is truncated” — it is not (starts at 0); “n = 7 is too small” — each point is a complete monthly total and the fall is exact arithmetic. Both are ruled out; the chart is not *wrong*, it is *incomplete*.

3 · Mechanism. The window runs from the annual peak (May) to near the annual trough (Nov), so it contains only the downhill half of a seasonal cycle. The identical May→Nov window gives -42.4% in 2023, -42.8% in 2024 and -41.9% in 2025 — every year. Meanwhile each of the 12 months of 2025 is above the same month of 2024 (**+5.9% to +13.3%**) and annual revenue grew 1,254.7 → 1,361.9 → 1,496.3 (+8.5%, +9.9%).

4 · Redesign. A cycle plot: one line per year, the analyst's window shaded. The same chart with the other years beside it.

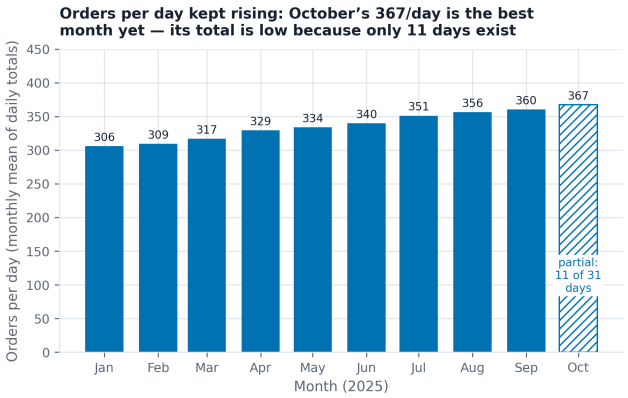
Case 6 · “Orders collapsed in October”

data: case06_monthly_partial.csv



SYNTHETIC course data, CDB601220 Week 4.

Rebuilt (the claim)



SYNTHETIC course data, CDB601220 Week 4. Hatched bar = incomplete month (1–11 Oct). Rate, not sum, because the periods are unequal.

Redesign (honest)

1 · Rebuild. Monthly totals with .sum(), bars from zero: ten bars; September 10,805, October 4,041 (37% of it) — self-check hit.

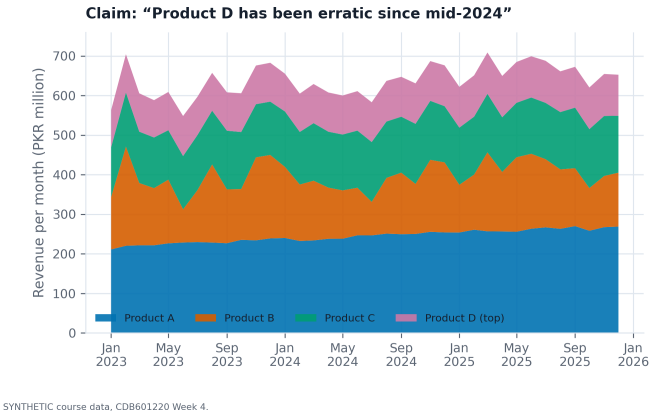
2 · Conclusion & decoy. Conclusion invited: a demand collapse. **Decoy:** “orders really were lost.” Ruled out: on the days that exist, October's daily orders ran 347–405, every day above January's *average* (306).

3 · Mechanism. The extract ends on **2025-10-11**: October has 11 of 31 days (35.5%). A sum compares an 11-day month with 30-day ones. Per day, October is **367.4** against September's 360.2 (+2.0%) — the highest of all ten months; prorated, October would be ≈11,388.

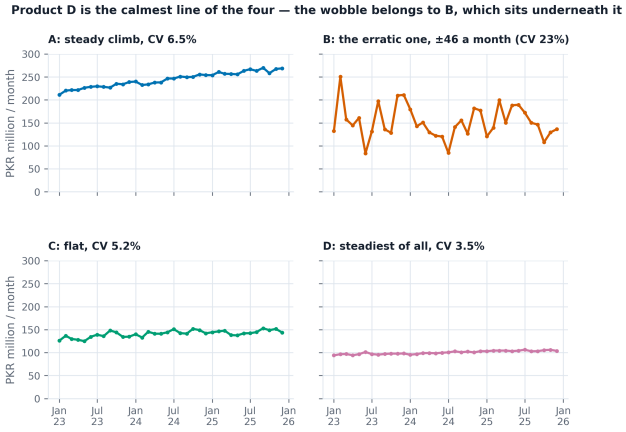
4 · Redesign. Orders per day (monthly mean) instead of a sum, with the incomplete month hatched and labelled “11 of 31 days”.

Case 7 · “Product D has been erratic since mid-2024”

data: case07_stacked_products.csv



SYNTHETIC course data, CDB601220 Week 4.



SYNTHETIC course data, CDB601220 Week 4. Same y-axis (0–300) in all four panels, so the swings are comparable.

Rebuilt (the claim) **Redesign (honest)**

1 · Rebuild. Stacked area, A at the bottom, D on top: the total envelope runs 548.3–708.7 — self-check hit.

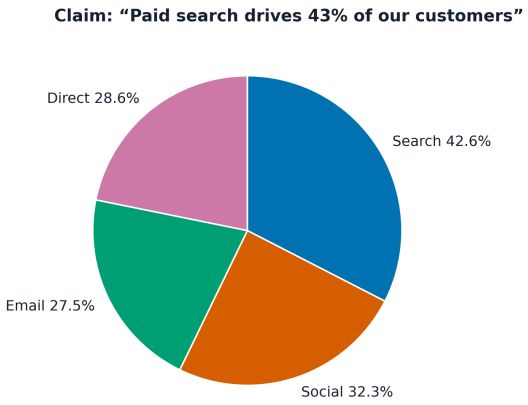
2 · Conclusion & decoy. Conclusion invited: D's revenue is volatile. **Decoy:** “D's own numbers are noisy.” Ruled out: D has the **lowest** coefficient of variation of the four (3.5%, against 23.4% for B), its month-to-month change has sd 2.0 (B: 45.8), its largest single move is 4.7, and it drifts calmly from 94.0 to 103.7.

3 · Mechanism. In a stacked area each layer's edges are the running total of everything beneath it. D sits on A+B+C, whose month-to-month sd is **47.5** — 24× D's own — and **product_b** (the column the recipe never isolates) accounts for 93% of that variance (correlation of B's changes with the baseline's: 0.98). The eye reads the wobble of D's *edges*, not its thickness.

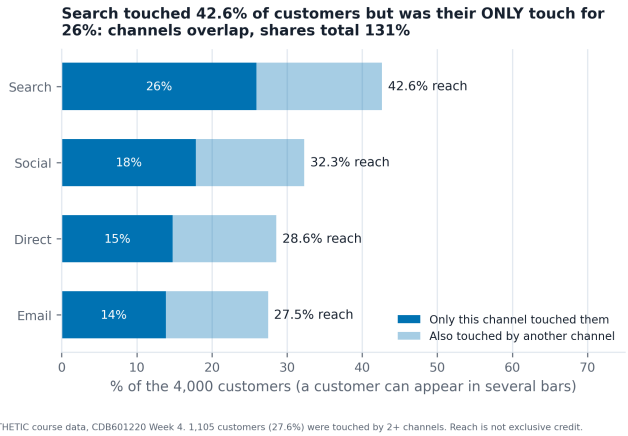
4 · Redesign. Small multiples, one panel per product on the same 0–300 axis, so the swing of each is directly comparable.

Case 8 · “Paid search drives 43% of our customers”

data: case08_channel_attrib.csv



SYNTHETIC course data, CDB601220 Week 4. Labels sum to 131.0%; matplotlib silently rescales the wedges to 100%.



SYNTHETIC course data, CDB601220 Week 4. 1,105 customers (27.6%) were touched by 2+ channels. Reach is not exclusive credit.

Rebuilt (the claim) **Redesign (honest)**

1 · Rebuild. Counted True per touch column over 4,000 customers: search 42.6%, social 32.3%, email 27.5%, direct 28.6% — self-check hit.

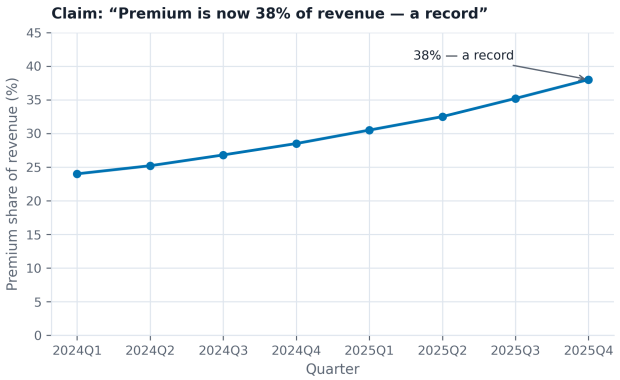
2 · Conclusion & decoy. Conclusion invited: search is responsible for 43% of customers. **Decoy:** “search is still the biggest, so the ranking at least holds.” As *reach* it does (42.6% vs 32.3%); but “drives” means credit, and search is the *only* touch for just 25.9% of customers (39% of those it touched were also touched by another channel).

3 · Mechanism. The four percentages add to **131.0%**. The parts are not exclusive: 1,105 customers (**27.6%**) were touched by two or more channels (2: 980, 3: 116, 4: 9), an average of 1.31 touches. A pie needs a partition of 100%. Worse, matplotlib silently rescales the wedges, so the “42.6%” wedge occupies only 32.5% of the circle while its label says 42.6%.

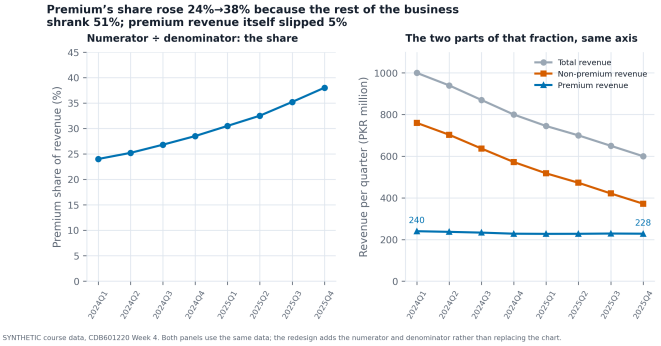
4 · Redesign. Horizontal bars of reach, each split into “only this channel” and “also another channel”; the legend and the title say they overlap.

Case 9 · “Premium is now 38% of revenue — a record”

data: case09_premium_share.csv



SYNTHETIC course data, CDB601220 Week 4.



SYNTHETIC course data, CDB601220 Week 4. Both panels use the same data; the redesign adds the numerator and denominator rather than replacing the chart.

Rebuilt (the claim) **Redesign (honest)**

1 · Rebuild. premium ÷ total per quarter, line, y from zero, last point annotated: 24.0% → 38.0% — self-check hit.

2 · Conclusion & decoy. Conclusion invited: premium is booming. **Decoy:** “the share is miscalculated.” Ruled out: 228.0 ÷ 600.0 = 38.0% exactly. The chart is arithmetically right — it is *incomplete*.

3 · Mechanism. The denominator collapsed: total revenue 1000 → 600 (**-40%**), non-premium 760 → 372 (-51%), while premium itself went 240 → 228 (**-5%**). Had the total stayed at 1000, the 2025Q4 share would be 22.8%. A rising share can be a shrinking company.

4 · Redesign. Not a different chart: the same share line with the numerator and denominator (premium, non-premium, total, in PKR m) beside it.